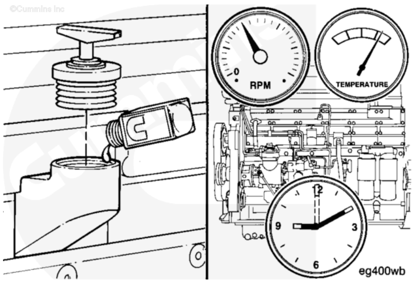
Trainings

Fluorescent Dye Tracer

Note : Shown in this procedure is the standard turbocharger. Although different than the water cooled turbocharger, the procedure remains the same.

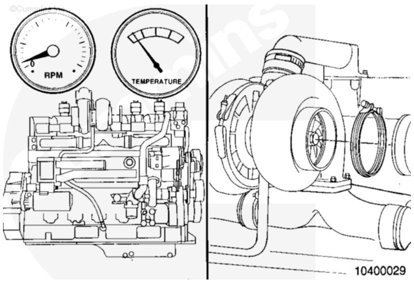
Add one unit of fluorescent tracer, Part number 3376891, to each 38 liters [10 U.S. gallons] of engine lubricating oil.

Operate the engine at low idle for 10 minutes.



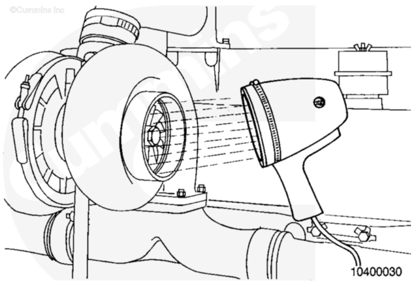
Shut the engine off.

Allow the turbocharger to cool and remove the exhaust pipe from the turbine housing.

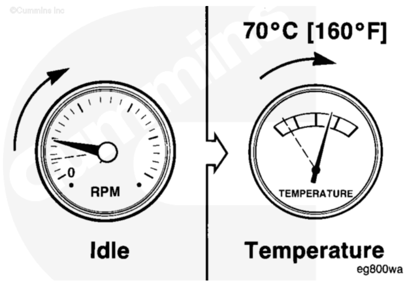


Use a high intensity black light to inspect the turbine housing outlet for oil.

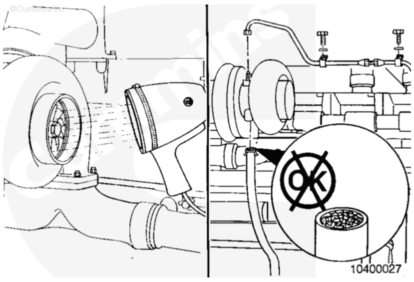
Note : A dark blue glow indicates fuel slobber, and a yellow glow indicates oil carryover.



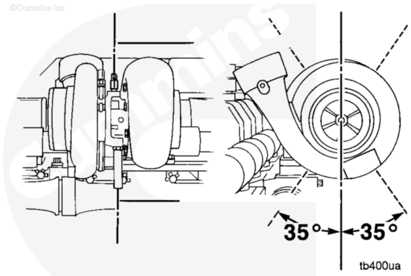
If fuel is found in the turbine housing, the problem is excess, unburned fuel. To correct the problem, decrease the idle period or increase the idle rpm. The coolant temperature needs to be maintained at a minimum of 71°C [160°F].



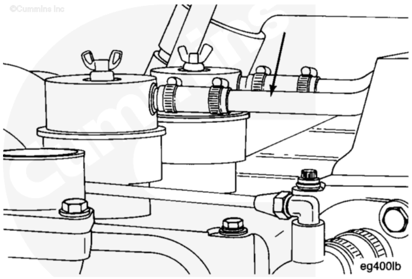
If oil is found in the turbine housing, remove the oil drain tube and check for restrictions. Clear any restrictions found.



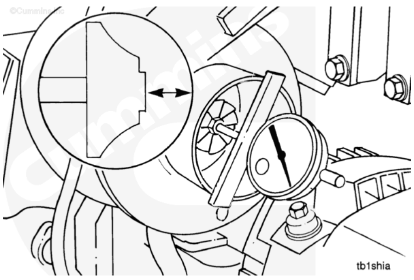
Check the angle of the drain tube. The angle of the tube **must** be within 35 degrees of vertical. Adjust the turbocharger, if necessary. Refer to Procedure 010-033 (/qs3/pubsys2/xml/en/procedures/20/20-010-033-tr.html).



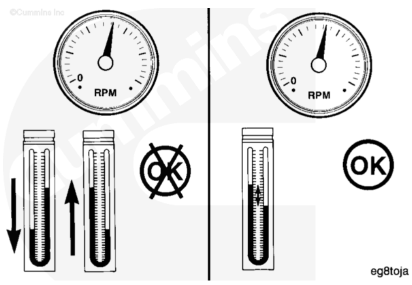
If the drain tube is free of restrictions and at the correct angle, check the crankcase breather and tube to be sure they are **not** plugged.



Check the turbocharger axial motion and radial clearance. Refer to Procedure 010-033 (/qs3/pubsys2/xml/en/procedures/20/20-010-033-tr.html).



If these checks do **not** reveal the problem, measure the crankcase pressure (blowby). Refer to Procedure 014-005 (/qs3/pubsys2/xml/en/procedures/20/20-014-005.html).



Last Modified: 05-Jul-2006